Fiscal Year:	FY 2009	**Task Last Updated:**	FY 08/10/2009
PI Name:	Hughes-Fulford, Millie Ph.D.		
Project Title:	Role of the Interleukin-2 Receptor in Signal Transduction		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Web Page:			
City:	San Francisco	**State:**	CA
Zip Code:	94121	**Congressional District:**	8
Comments:	PI is deceased as of February 2021		
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-04
Start Date:	01/15/1999	**End Date:**	03/31/2009
No. of Post Docs:	3	**No. of PhD Degrees:**	1
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Soyuz/ISS		
Flight Assignment:	ISS (changed 7/07 per PI) Soyuz (changed 7/07 per PI) NOTE: Received NCE to 3/31/2009 (from 9/30/2008) per A. Chu/ARC (9/08)		
Key Personnel Changes/Previous PI:	Augusto Cogoli retired and the ESA PI is now Isabelle Walthers. The flight experiment on Soyuz was executed in orbit by ESA Astronaut Thomas Reiter. The incubation time was at 1.5 hours instead of the planned 4 hours. As a result, we had to make standards for the new timepoint.		
COI Name (Institution):	Walther, Isabelle (Swiss Federal Institute of Technology)		
Grant/Contract No.:	NAG2-1286		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This work is collaboration between Professor Augusto Cogoli and Professor Millie Hughes-Fulford to study the role of the interleukin-2 receptor in signal transduction of T cells. The nearly total loss of T cell activation in real microgravity was discovered in a Spacelab in 1983 and was confirmed later by several experiments in space. Inhibition of IL-2R expression is probably one of the causes of such loss of activation. It is planned to analyze the pathway of mitogenic signal transduction in T cells in true microgravity as well as at variable G levels between 0 and 1.g. The objectives of this research project are: 1. To investigate selected critical steps of T cell activation 2. To test the hypothesis that a failure in the expression of the IL-2R is causing the loss of activity in microgravity 3. To identify windows of sensitivity to gravity during the activation process 4. To establish thresholds of gravi-sensing T cells. The main experimental approach will consist of the activation of cultures of purified peripheral blood cells with T cell mitogens in real and simulated microgravity. The specific IL-2R mRNA will be quantitatively determined with the reverse Transcriptase-polymerase chain reaction (RT-PCR) technology, the insertion of IL-2R in the membrane will be visualized by immunofluorscence and its secretion in the supernatant will be measured by immunoassay. Signal transduction will be furthered analyzed by determining the early expression of oncogenes. The Laboratory of Cell Growth (LCG), headed by Dr. Millie Hughes-Fulford 's lab will analyse the gene expression in T cells and determine which genes require gravity for full induction..
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	All terrestrial life began in a gravity field. This study of alterations of early T-cell activation microgravity gives us the unique opportunity to examine the role of Earth's gravity in immune function. In the late 60's returning Apollo astronauts were found to have reduced immune function after spaceflight, taking approximately 7 days to recover normal function. Later studies by Dr. Cogoli demonstrated changes in the T-cells during exposure to microgravity. This program will examine the expression of 30,000 genes and 10,00 ESTs including IL-2, and the three IL-2 receptor subunits (alpha, beta and gamma) in microgravity with and without an artificial 1g gravity vector and compare those results with ground controls. Results will be verified using qRTPCR. These studies show the regulation of the earliest signals that cause the T-cell to activate and the role of normal earth's gravity in that signaling. These results may provide new pharmaceutical targets for the treatment of immunosuppression.
Task Progress:	Early Apollo missions showed that the crews returned with a supressed immune system. Here, we examined the differences in molecular function of activation of T cells in 1.g and µg. The molecular functions of a gene product are the jobs that it does or the "abilities" that it has. These may include transporting things around, binding to things, holding things together and changing one thing into another. This is different from the biological processes the gene product is involved in, which involve more than one activity. One way to understand this is to consider the analogy of a company or organization. Individuals (gene products) have different abilities or tasks (functions) and they work together to achieve different goals (processes). It is easy to confuse a job title (gene product name) with a function; for example, 'secretarial activity' may seem like a valid function because you have a good conceptual idea of what a secretary does. However, in different companies, secretaries might do different things. One secretary might have the functions 'typing', 'answering phone' and 'making coffee', while another might have these functions and additionally 'photocopying'. In the Gene Ontology, a function should be unambiguous and it should mean the same thing no matter what species you are dealing with. (see http://www.geneontology.org/). Hardware: Cells were grown in Lysis hardware. Cells were placed into the hardware in Baikonur, Kazakhstan prior to the launch of the Soyuz taxi flight to the International Space Station (ISS). Once on ISS, the cells were placed into the KUBIC incubator for 3 hours prior to activation. Cells were then activated and placed into 1.g centrifuge or µg conditions inside the incubator. METHODS: RNA Cleanup and processing Cell total RNA was stabilized by RNAlater (Qiagen, Valencia, California). RNA was isolated using Trizol® followed by RNeasy® Micro kit (Qiagen, Valencia, California). RNA concentration and purity was determined from measuring absorbance at 260 nm and 280 nm and 0.3 µg total RNA was run on a 1% denaturing gel, or 100 ng total RNA was loaded on the 2100 Bioanalyzer (Agilent, Palo Alto, California) to verify RNA integrity. UPDATE OF LEUKIN FLIGHT DATA: The re-flight of Leukin experiment on the International Space Station (ISS) transported on the Soyuz Taxi resulted in some problems of our primary objective to examine the expression of IL-2Ralpha since the experiment was prematurely terminated two and a half hours earlier than required. The experiment was further affected by the temperature excursions prior to launch of the liftoff resulting in the low yield of RNA in the Lysis hardware. The MIA hardware was made of metal which conducts heat/cold efficiently resulting in lowered number of cells that survived the launch and flight. The reduction of cells to the average of ~50,000 to 100,000 cells per well, a significant reduction to the 1 million cells that were originally placed into the containers. The near frozen state of the MIA hardware after the operation made the cells unusable for the planned morphology studies post flight. Amount of RNA insufficient for analysis: The premature termination of the Lysis hardware experiment of Leukin resulted in a significantly reduced amounts of RNA. At the time of return of the samples, we needed a full 1-2ug of RNA for the gene array and 1.5ug of RNA for 6 real-time qRTPCR reactions (a total of 3.5 ug of RNA needed for each samples in each condition), making meaningful analysis of the RNA impossible. In the intervening time, we worked on the technology of Gene arrays and real-time qRTPCR to reduce the amount of RNA required for analysis. After many months of work, we were able to analyze the gene arrays with a reduced amount of RNA making analysis possible. We then had to secure additional funding to cover new ground controls for the 1.5 hour time point and the new technology for the gene arrays. Significant changes in gene expression of microgravity when compared to 1.g flight data. Since funding was secured, for the new post-flight analysis of Leukin, we have made some interesting discoveries about the effect of microgravity on early T-cell activation.

	Data from the 1.5 hour flight experiment shows that Gene Ontology (GO) categories of gene expression are changed in the microg samples, when comparing 1g expression with microgravity gene expression. First, we found that in early activation the differences in microgravity were largely due to binding and changes in signal transduction and regulation of transcription. This was expected since the early portion of T-cell activation involves binding of the activators and the activation of signal transduction of gene expression (transcription). The data used for comparison for the flight data was taken from ground based studies using the Random Positioning Machine (RPM) to simulate microgravity. 143 genes are expressed in 1.g flight conditions and that are not expressed in microgravity. Preliminary examination of the GO differential gene expression shows that there is upregulation of 143 genes in the 1.g flight samples and these genes are not upregulated in the flight microgravity controls. Detailed Analysis are nearing completion: We have examined the expression of 45,000 genes/and ESTs from the triplicate independent biological samples from each of the conditions (n=12), not an easy matter when considering that each of the 12 arrays has approximately 2,000,000 hybridization spots. One manuscript is nearing completion for submission, a second is waiting for NASA to examine patent disclosure for potential patent application as a result of our unique findings.
Bibliography Type:	Description: (Last Updated: 03/10/2018)
Articles in Peer-reviewed Journals	Hughes-Fulford M, Chang TT, Li C-F. "Effect of Gravity on Monocyte Differentiation." Journal of Gravitational Physiology, In Press, August 2009. , Aug-2009
Articles in Peer-reviewed Journals	Hughes-Fulford M, Li CF. "The role of FGF-2 and BMP-2 in regulation of gene induction, cell proliferation and mineralization." J Orthop Surg Res. 2011 Feb 9;6:8. [Epub ahead of print] PMID: 21306641 , Feb-2011
Books/Book Chapters	Hughes-Fulford M, Boonstra JL. "Cell Mechanotransduction: cytoskeleton and related signaling pathways." in "Cell Mechanochemistry. Biological Systems and Factors Inducing Mechanical Stress, Such as Light, Pressure and Gravity." Ed. M. Monici, J. van Loon. Kerala, India : Transworld Research Network, 2010. p. 75-96., Jun-2010